

COMS20017 – Algorithms & Data

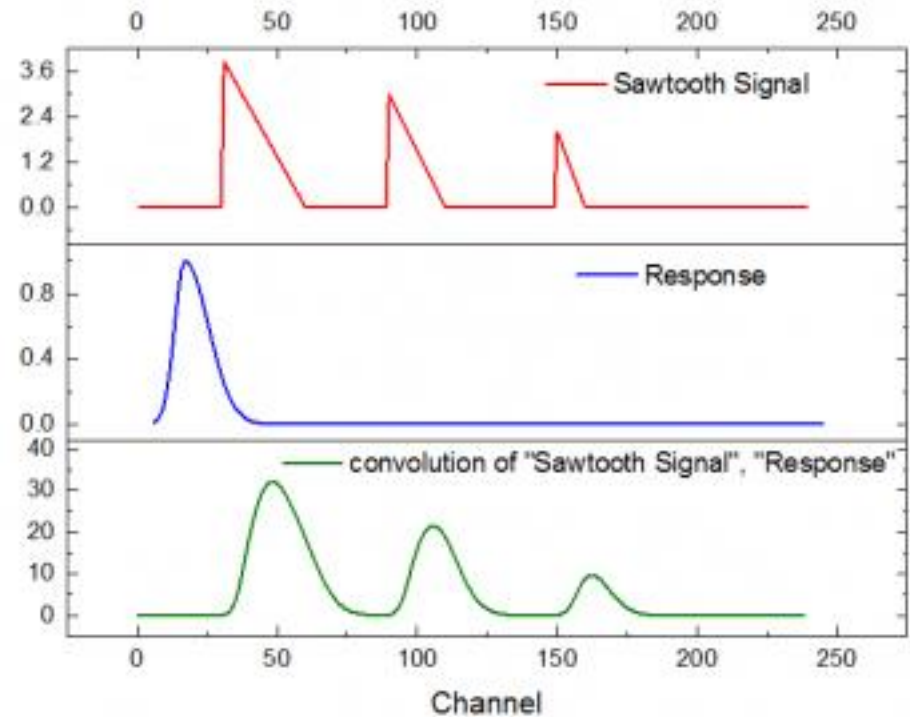
3_0	3_1	2_2	1	0
0_2	0_2	1_0	3	1
3_0	1_1	2_2	2	3
2	0	0	2	2
2	0	0	0	1

12.0	12.0	17.0
10.0	17.0	19.0
9.0	6.0	14.0

Convolutions

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Next in Data



Feature Selection and Extraction

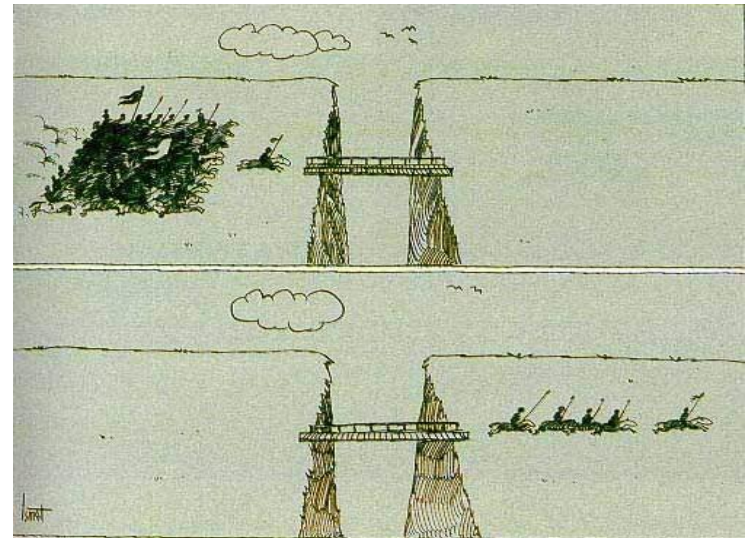
- Signal basics and Fourier Series
- 1D and 2D Fourier Transform
- Another look at features
- PCA for dimensionality reduction
- **Convolutions**

Spatial Filtering

We can filter signals and symbols in the spatial/time domain:

- introduce some form of enhancement
 - remove noise/outliers
 - smoothing/averaging out detail
 - sharpening/highlighting detail
- prepare for next processing stage
 - feature extraction

Filters are also referred to as
kernels or *masks*.



Spatial Filtering

Many spatial filters are implemented with **convolution** masks.

To do convolution, we need to know about **neighbourhoods**.

Symbolic Data

ATAGACATGGC



neighbours of G?

1D signal data



2D signal data

3	2	4	4	2	6
3	4	5	4	3	6
4	2	5	4	3	3
3	0	4	1	2	6
3	2	4	5	2	6

.....

Convolution mask is applied to each signal sample and its neighbourhood.

Convolution

$$g(x) = h(x) * f(x)$$

Convolution
Kernel $h(x)$

1	2	1
---	---	---

Signal
 $f(x)$

1	1	2	2	1	1	0	1	2	2
---	---	---	---	---	---	---	---	---	---



Filter
Response
 $g(x)$

	<u>5</u> 4								
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Convolution

$$g(x) = h(x) * f(x)$$

Convolution
Kernel $h(x)$

1	2	1	1	2	1	2	1	2	1	2	1
---	---	---	---	---	---	---	---	---	---	---	---

Signal
 $f(x)$

1	1	2	2	1	1	0	1	2	2
---	---	---	---	---	---	---	---	---	---

Filter
Response
 $g(x)$

	↓	↓	↓	↓	↓	↓	↓	↓	↓
☹	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{7}{4}$	$\frac{5}{4}$	$\frac{3}{4}$	$\frac{2}{4}$	$\frac{4}{4}$	$\frac{7}{4}$	☹

Convolution

➤ f is the signal, h is the convolution filter

➤ h has an origin

Example 1D kernel $\frac{1}{5}$

-1	3	-1
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➤ Normalization factor (sum of the absolute values of the filter) is also part of the filter.

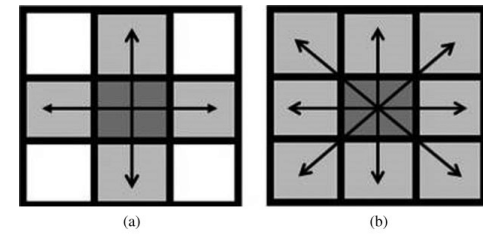
$$g = f(t) * h(t)$$

$$g = \int_{-\infty}^{\infty} f(t - \tau)h(\tau)d\tau = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$

➤ The discrete version of convolution is defined as:

$$g(x) = \sum_{m=-s}^s f(x - m)h(m) \quad \text{for } s \geq 1$$

2D Spatial Filtering - Connectivity



Determine the connectivity for *neighbourhoods*:

2D signal data

4-connectivity

3	2	4	4	2	6
3	4	5	4	3	6
4	2	5	4	3	3
3	0	4	1	2	6
3	2	4	5	2	6

8-connectivity.

3	2	4	4	2	6
3	4	5	4	3	6
4	2	5	4	3	3
3	0	4	1	2	6
3	2	4	5	2	6

2D Convolution

- The discrete version of 2D convolution is defined as

$$g(x, y) = \sum_{m=-1}^1 \sum_{n=-1}^1 f(x-m, y-n)h(m, n)$$

Shorthand form:

$$g = f * h$$

Convolution symbol

h					
	-1	0	1		
-1	-1	0	1		
0	-2	0	2	$x-1$	
1	-1	0	1	x	
				$x+1$	

			f		
			$y-1$	y	$y+1$
			43	12	61
			44	45	60
			43	50	61

➡ -68

$$\begin{aligned}
 & f(x+1, y+1)h(-1, -1) \\
 & + f(x+1, y)h(-1, 0) \\
 & + f(x+1, y-1)h(-1, 1) \\
 & + f(x, y+1)h(0, -1) \\
 & + f(x, y)h(0, 0) \\
 & + f(x, y-1)h(0, 1) \\
 & + f(x-1, y+1)h(1, -1) \\
 & + f(x-1, y)h(1, 0) \\
 & + f(x-1, y-1)h(1, 1)
 \end{aligned}$$

2D Correlation

- The discrete version of 2D correlation is defined as

$$g(x, y) = \sum_{m=-1}^1 \sum_{n=-1}^1 f(x+m, y+n)h(m, n)$$

h			f				
	-1	0	1		$y-1$	y	$y+1$
-1	-1	0	1				
0	-2	0	2	$x-1$	43	12	61
1	-1	0	1	x	44	45	60
				$x+1$	43	50	61

➡ 68

Correlation=Convolution
when kernel is symmetric
under 180° rotation, e.g.

1	2	1
---	---	---

Spatial Filtering using 2D Convolution (actually Correlation)

Simple example

f

3 ₀	3 ₁	2 ₂	1	0
0 ₂	0 ₂	1 ₀	3	1
3 ₀	1 ₁	2 ₂	2	3
2	0	0	2	2
2	0	0	0	1

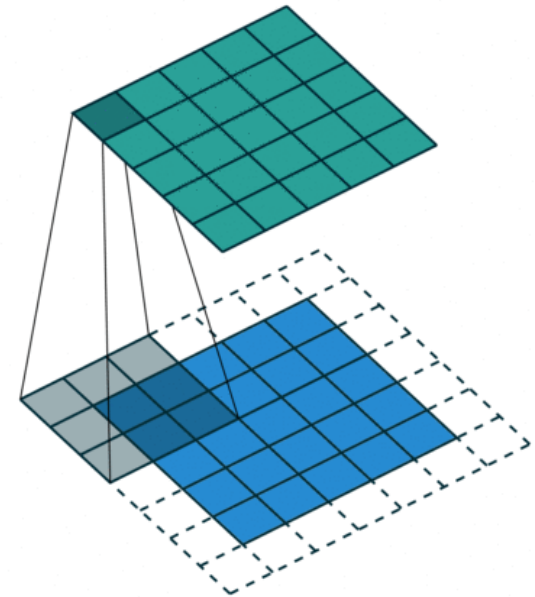
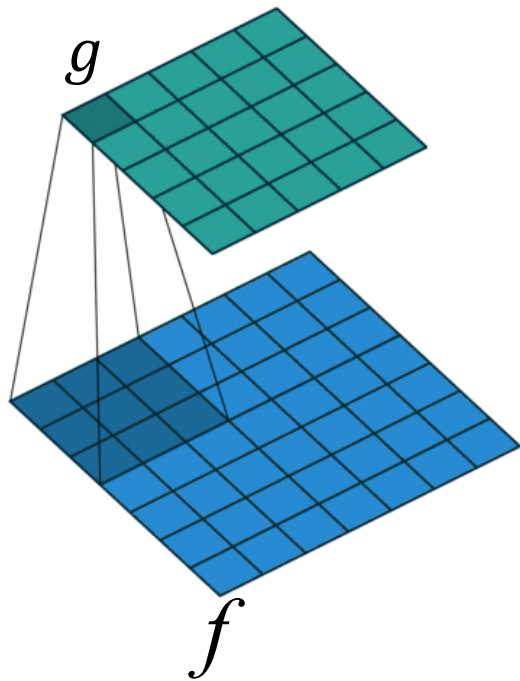
g

12.0	12.0	17.0
10.0	17.0	19.0
9.0	6.0	14.0

h

0	1	2
2	2	0
0	1	2

Use padding for same size result



Example: Spatial Low/High Pass Filtering

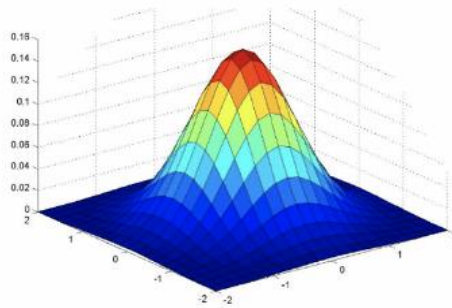
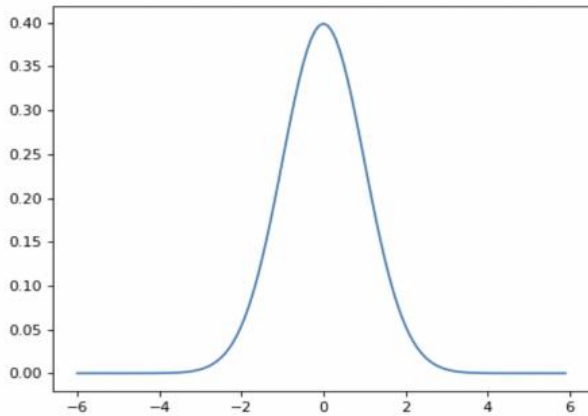
- 1D: turning the treble/bass knob down on audio equipment!
- 2D: smooth/sharpen image

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\frac{1}{16} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

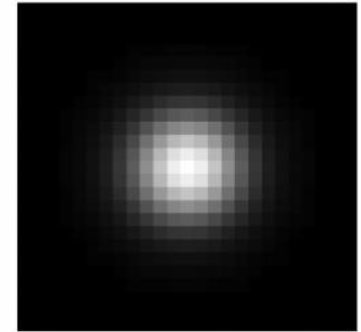


Gaussian Low Pass Filter



Gaussian kernel

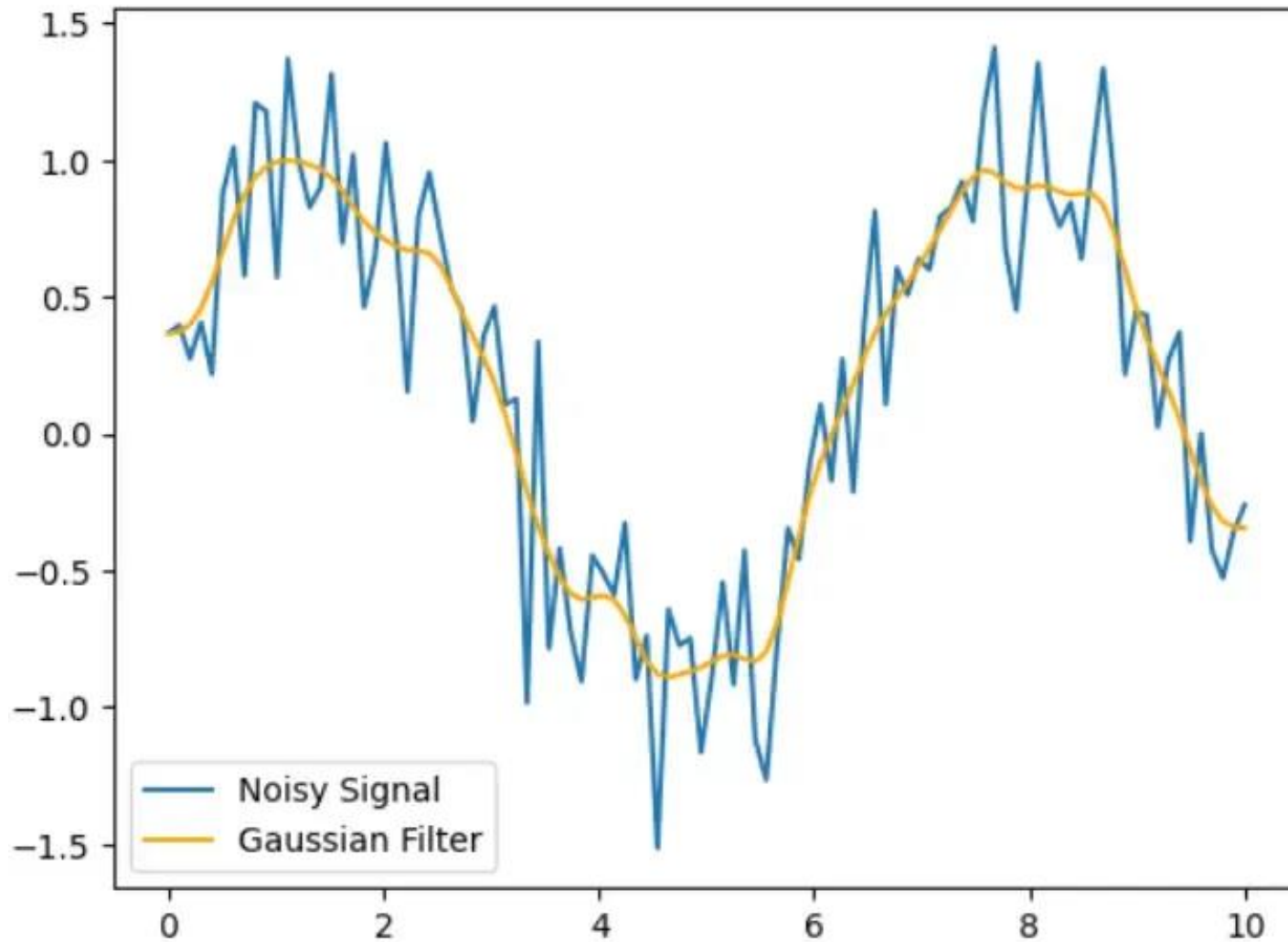
$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



Example:

$$\frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix} = \frac{1}{16} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \end{pmatrix}$$

Example: Gaussian Low Pass Filter in 1D



Example: Edge Features

- Edges occur in images where there is discontinuity (or change) in the intensity function.
- Gradient points in the direction of most rapid change in intensity
- Biggest change → derivative has maximum magnitude.

$$f(x, y)$$

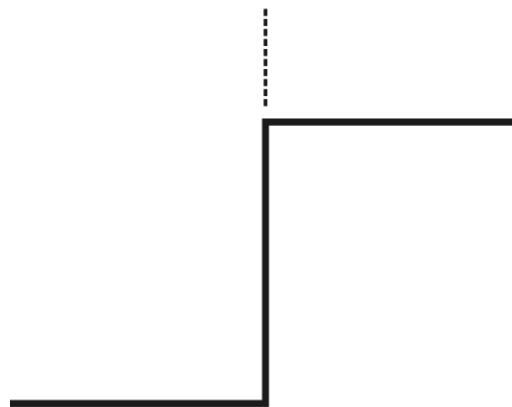


$$\|\nabla f\| = \sqrt{\left[\frac{\partial f}{\partial x}\right]^2 + \left[\frac{\partial f}{\partial y}\right]^2}$$

Edge detection: basic concept

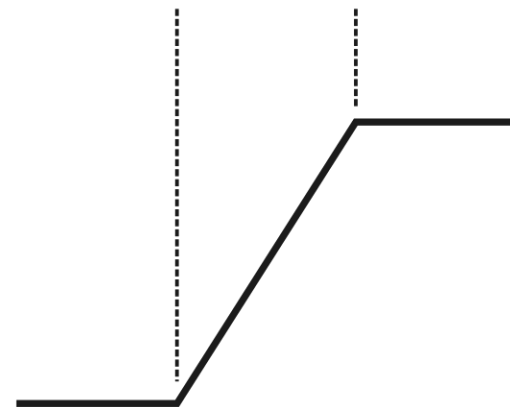
- Edge: a boundary between two image regions having distinct characteristics according to some feature (e.g., gray level, color, or texture).
- In grayscale 2D images: a significant variation of the intensity function across a portion of the image.

Model of an ideal digital edge



Gray level profile of a horizontal line through the image

Model of a ramp digital edge



Gray level profile of a horizontal line through the image

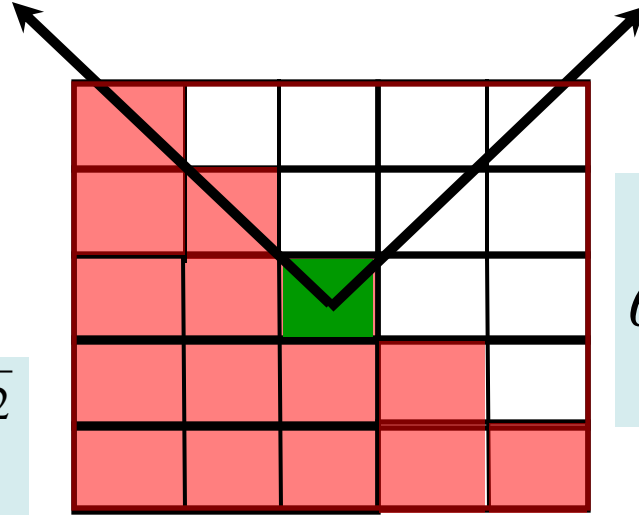
Edge Measures

Edge direction

$$\phi = \theta - \frac{\pi}{2}$$

Edge magnitude

$$\|\nabla f\| = \sqrt{\left[\frac{\partial f}{\partial x}\right]^2 + \left[\frac{\partial f}{\partial y}\right]^2}$$



gradient direction

$$\theta = \tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right)$$

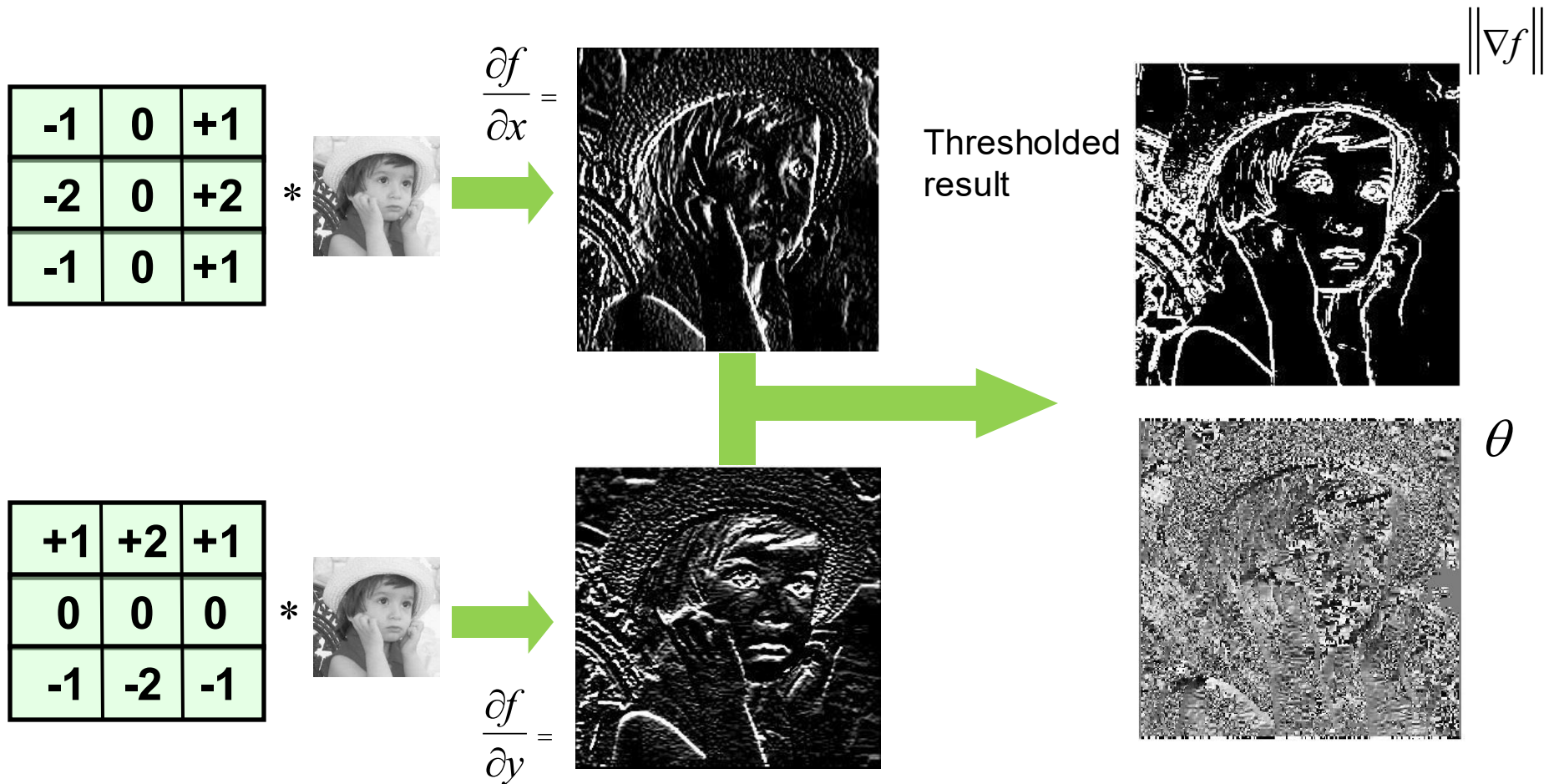
$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ 0 \end{bmatrix}$$

$$\nabla f = \begin{bmatrix} 0 \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

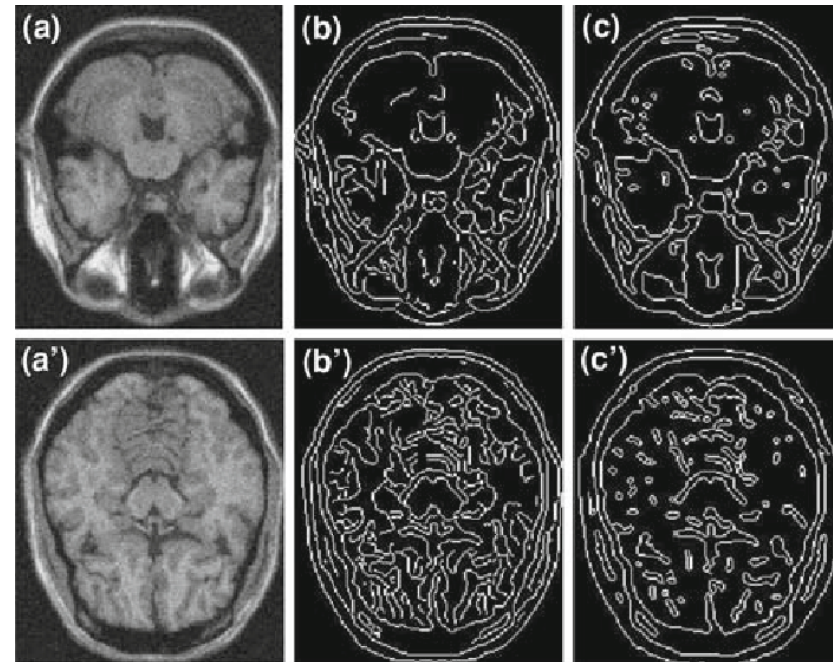
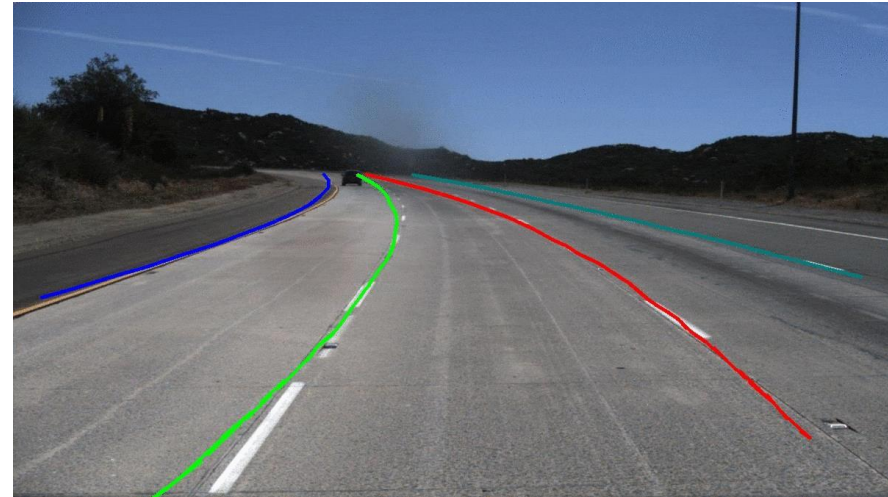
$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Sobel Edge Detector

- 2D gradient measurement in two different directions.
- Uses these 3x3 convolution masks:

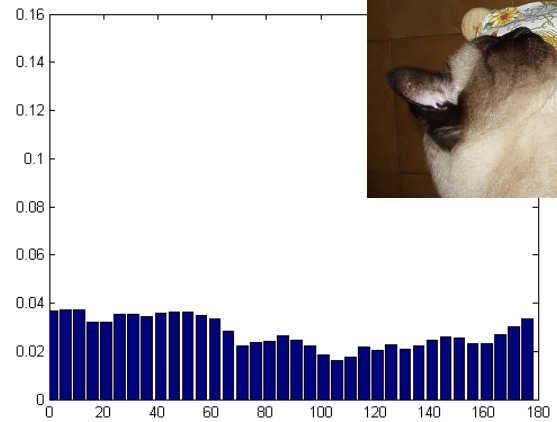
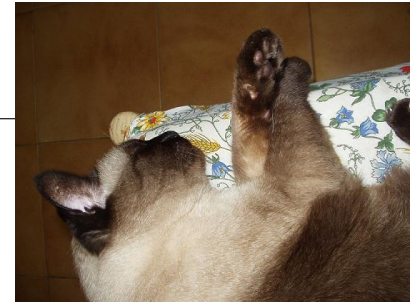
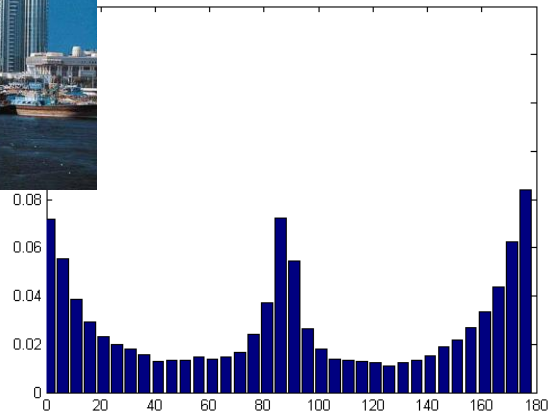
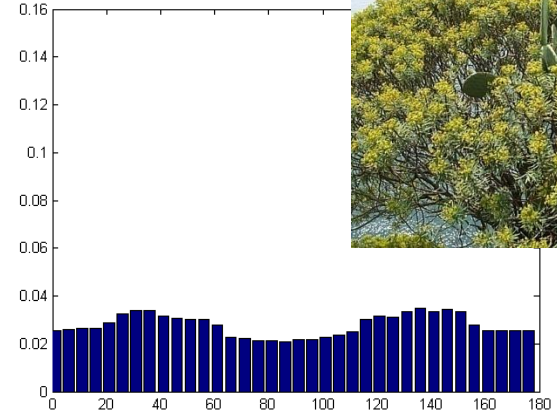
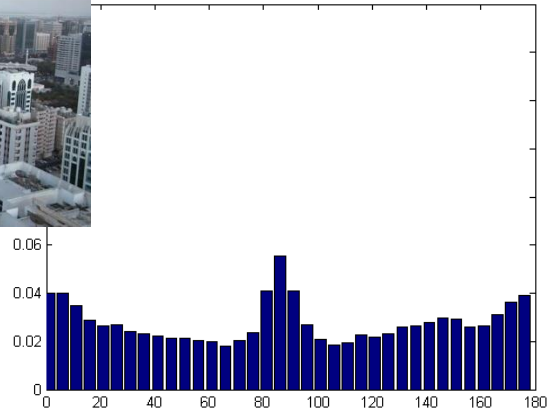


Examples



Histogram of Edge Gradient Directions

$$\theta = \tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right)$$



Matlab: Sobel Edge Detection

```
% Sobel edge detection
A = imread('romina.gif');
fx = [-1 0 1; -2 0 2; -1 0 1]
fy = [1 2 1; 0 0 0; -1 -2 -1]
gx = conv2(double(A),double(fx))/8;
gy = conv2(double(A),double(fy))/8;
mag = sqrt((gx).^2+(gy).^2);
ang = atan(gy./gx);
figure; imagesc(mag); axis off; colormap gray
figure; imagesc(ang); axis off; colormap gray
```

See unit github page for code in Python.

Spatial/Frequency Domain Filtering

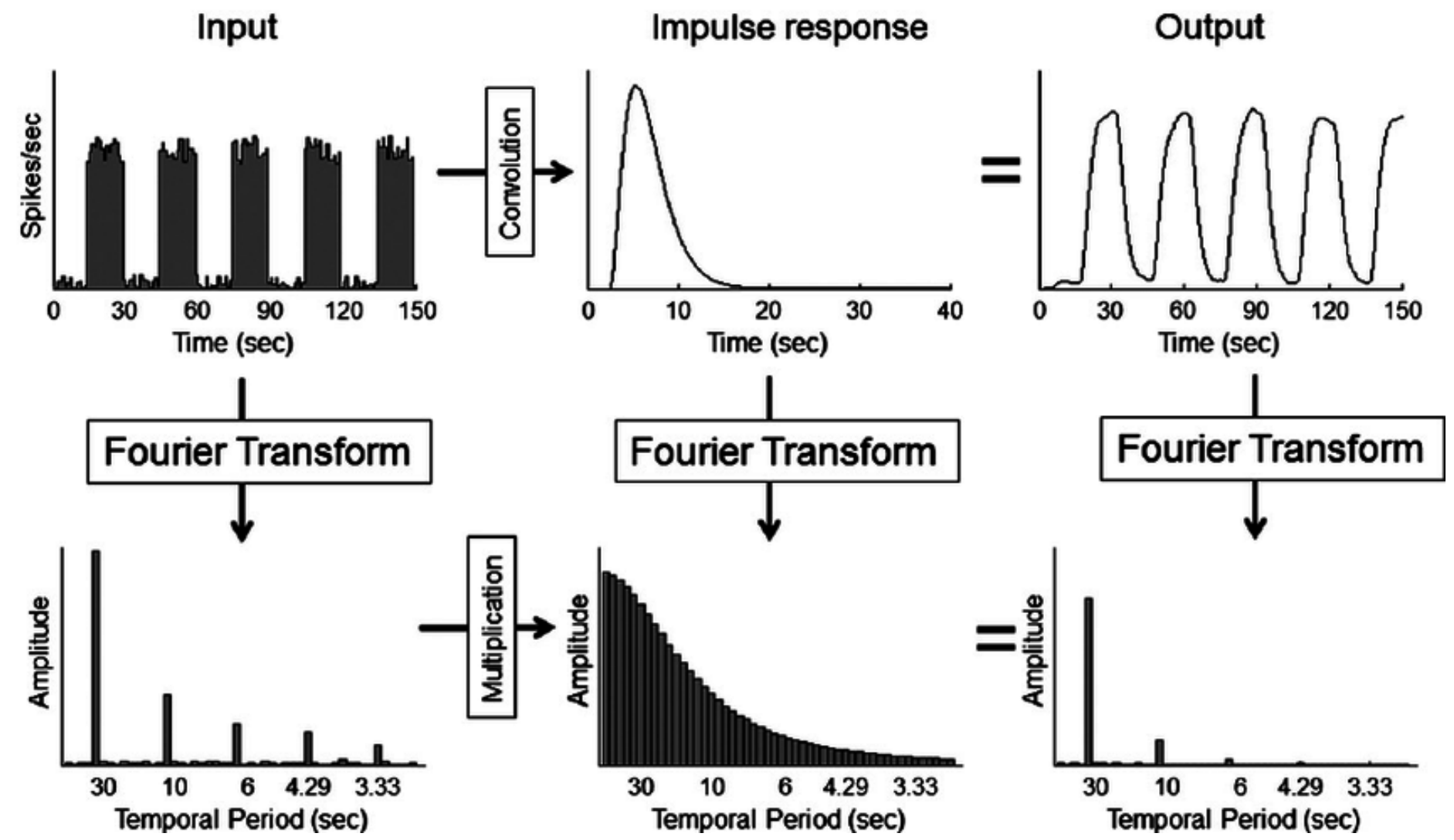
➤ Convolution Theorem:

Convolution in the spatial domain
is equivalent to
multiplication in the frequency domain
and vice versa

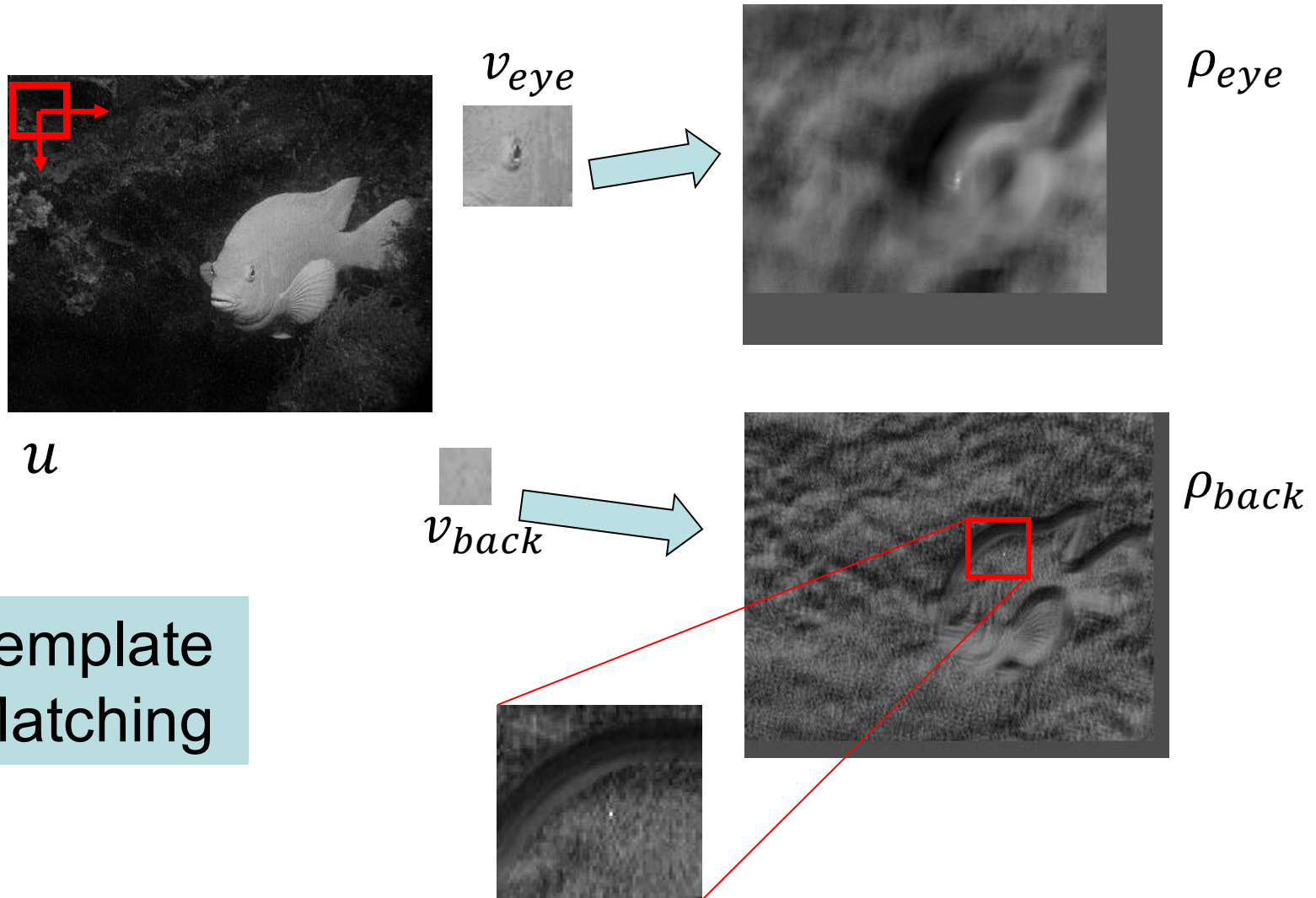
$$g(x) = f(x) * h(x) \quad \Longleftrightarrow \quad G(u) = F(u) H(u)$$

$$g(x) = f(x) h(x) \quad \Longleftrightarrow \quad G(u) = F(u) * H(u)$$

Example: Convolution in SD is Multiplication in FD



Correlation Example: Fish parts!



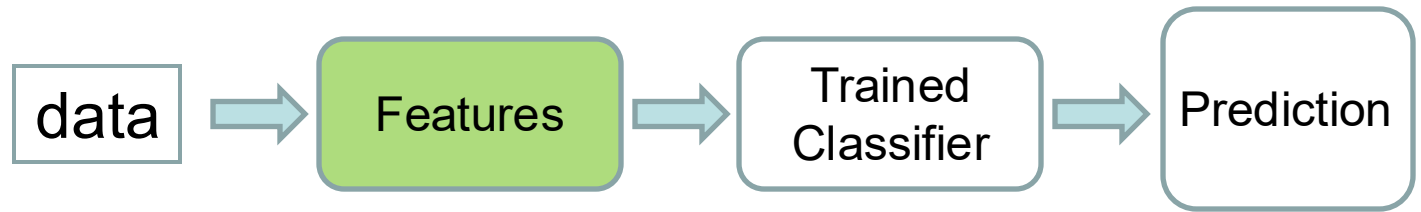
Convolution Exercise

$$P = \begin{pmatrix} -1 & 4 & -1 \\ 4 & 10 & 4 \\ -1 & 4 & -1 \end{pmatrix} \quad Q = \begin{pmatrix} 5 & 3 & 0 & 0 & 7 \\ 3 & 3 & 2 & 1 & 4 \\ 6 & 5 & 1 & 1 & 6 \\ 6 & 3 & 4 & 1 & 4 \\ 4 & 4 & 5 & 0 & 4 \end{pmatrix}$$

Apply convolution filter P to matrix Q at location (2,3).

Answer = ?

Well Done for Lasting This Far!



Feature Selection and Extraction

- Signal basics and Fourier Series
- 1D and 2D Fourier Transform
- Another look at features
- PCA for dimensionality reduction
- Convolutions